

Revised Generation Interconnection System Impact Study Report

For Queue Project Q19-1174

Coos Bay 138 kV
October 02, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1174. The Interconnection Customer proposes a 239.5 MW solar photovoltaic facility interconnecting at the Coos Bay 138 kV substation in the West Virginia area. The request was received on March 24, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 377 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$19,507,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Station 54 to Toone 138kV
Contingency: Aspen to Crandall 345kV
Loading: 132.5% (Rating: 266 MVA)
DFAX: 41.2%
Pre-project loading: 95.5%
2. Facility: Toone to N Omaha 138kV
Contingency: Crandall to Aspen 345kV L
Loading: 123.9% (Rating: 292 MVA)
DFAX: 41.2%
Pre-project loading: 90.1%
3. Facility: Sta 54 to Downtown 138kV
Contingency: Weyers Cave Substation to Greenbank 138kV
Loading: 115.8% (Rating: 218 MVA)
DFAX: 10.4%
Pre-project loading: 104.4%
4. Facility: N Omaha to Coos Bay 138kV
Contingency: Crandall to Aspen 345kV
Loading: 114.2% (Rating: 254 MVA)
DFAX: 53.1%
Pre-project loading: 64.2%
5. Facility: Kahe 2 to Fish Crk 138 138kV
Contingency: Greenland Gap to Santee Cooper 345kV
Loading: 109.2% (Rating: 356 MVA)
DFAX: 9.0%
Pre-project loading: 103.1%

6. Facility: W Loop to North River D.P 138kV
Contingency: Weyers Cave to Greenbank 138kV
Loading: 103.6% (Rating: 241 MVA)
DFAX: 10.4%
Pre-project loading: 93.3%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Station 54 to Toone 138kV	Rebuild line with higher-capacity conductor	\$3,486,000	56% (\$1,952,000)
Toone to N Omaha 138kV	Rebuild line with higher-capacity conductor	\$7,251,000	56% (\$4,060,000)
Sta 54 to Downtown 138kV Line	Rebuild line with higher-capacity conductor	\$5,735,000	61% (\$3,496,000)
N Omaha to Coos Bay 138kV	Rebuild line with higher-capacity conductor	\$10,967,000	70% (\$7,723,000)
Fish Crk 138 to Kahe 2 138kV Line	Reconductor line	\$1,763,000	87% (\$1,541,000)
W Loop to North River D.P 138kV	Reconductor line	\$1,206,000	61% (\$735,000)
	Total allocated to project		\$19,507,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.